

1. What are the three main layers of software architecture? Write one line about each.

- **Presentation Layer (UI):** The visual interface where the user interacts with the application (e.g., screens, buttons, and layouts).
- **Application Layer (Business Logic):** The underlying system that processes the user's commands, applies rules, and handles the core functionality of the app.
- **Data Layer (Database):** The storage system where the application securely saves, retrieves, and updates all necessary information.

2. Define client and server in your own words with one real-life example.

Client: A device or application (like your laptop or web browser) that requests a specific service, resource, or piece of data.

- **Server:** A powerful, centralized computer or system that listens for these requests, processes them, and sends the data back to the client.
- **Real-life Example:** Think of dining at a restaurant. You are the **client** who looks at the menu and requests a meal. The kitchen is the **server** that receives your order, prepares the food, and brings it out to you.

3. Write 2 differences each between desktop apps, web apps, and mobile apps.

- **Desktop Apps:**
 1. They must be installed directly onto a computer's local hard drive.
 2. They can frequently be used completely offline without an internet connection.
- **Web Apps:**
 1. They do not require installation and are accessed entirely through a web browser (like Chrome or Safari).
 2. They rely on an active internet connection to function properly.
- **Mobile Apps:**
 1. They are custom-built for the specific operating systems and touch interfaces of smartphones and tablets.
 2. They are typically downloaded and updated through centralized platforms like the Apple App Store or Google Play Store.

4. Name any 3 databases and mention whether each is SQL or NoSQL.

- **MySQL:** SQL (Relational Database)
- **MongoDB:** NoSQL (Non-Relational Database)
- **PostgreSQL:** SQL (Relational Database)